DATA STRUCTURES

LAB NO. 10

TASK NO. 02

NAME : ALEENA AKHTAR

SAP ID : 69273

SUBMITTED TO : MISS NEHAN SAHAR

DATE : 27TH APRIL, 2026

Riphah International University

Gulberg Green Campus

‘CIRCULAR QUEUE’

1. Queue Full Condition

A queue is considered **full** when there is no more space available to insert a new element. In an array-based queue, memory is fixed in size, so once all positions are occupied, no further elements can be added.

The condition for a full queue is:

rear == size - 1

This means that the rear pointer has reached the last index of the array. Since arrays have a fixed size, even if there are empty spaces at the beginning (due to dequeue operations), a simple linear queue cannot reuse them. As a result, the queue is treated as full.

In simple words:
When the queue has reached its maximum capacity and cannot accept more elements, it is called full.

2. Queue Empty Condition

A queue is said to be **empty** when it contains no elements. In this state, no deletion (deQueue) operation can be performed because there is nothing to remove.

The condition for an empty queue is:

front == -1 OR front > rear

- front == -1 means the queue has not been initialized or all elements have been removed.
- front > rear occurs after multiple dequeue operations, indicating that all elements have already been deleted.

In simple words:
When there are no elements left in the queue, it is called empty.

THANK YOU!!